

Discussion Module:

Capacity Planning and Tuning

Discussion/Information Only-

Anything less than (n) nodes and (mGB) data can produce spurious results. A capacity planning and tuning lab exists in the advanced class; 14 + nodes, plus driver nodes, production sized data, other.

- Topics related to capacity planning
- Topics related to tuning
- DSE Search specific only

Discussion Lab:

Discuss



Background:



- Fictitious company operates out of Miami, with stand alone production servers/data centers in all/most countries in S.America and LATAM.
- Shared development/test server; 24 cores, 84 TB disk
- Explosive growth over 7-10 years-
 - Production servers in Brazil, Mexico become 12x to 48x larger than development/test
 - No real HA disaster planning/test. Best estimate is to serve a (failed) country out of an adjacent country/server
 - Highly spikey system load; imagine pay per view on World Cup Soccer style events
- Too much unplanned downtime

Today 1:31 PM

When I give you this patch, what is your level of comfort that this will solve the production problem ?

Today 2:34 PM

Zero, In fact, you've given us patches before that have brought production down, and that can't happen again.

The last time you did (recovery procedure x); How long did that take ?

Today 4:17 PM

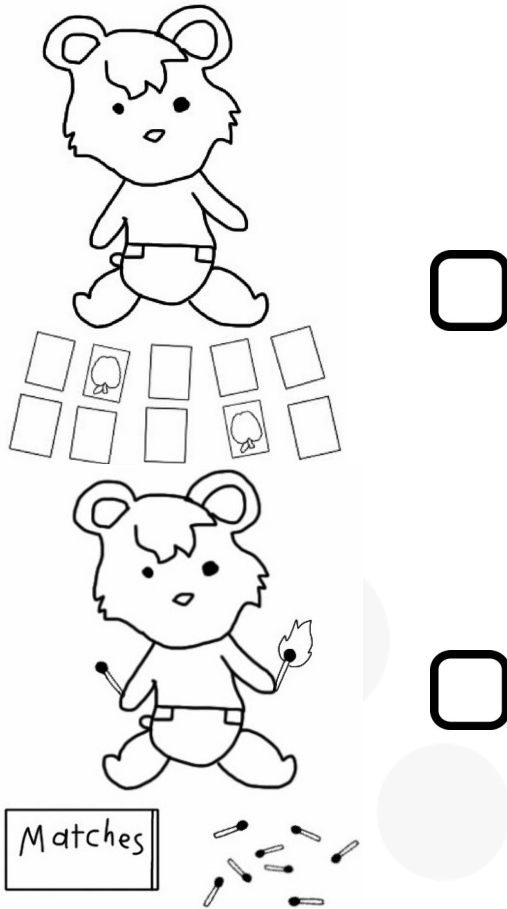
We haven't done that in 6-8 years. Since that time production volumes have gone up 48x. This can't take more than (n) time, and it can't affect production.

Vendor <- -> Customer



www.snoopy.com

Best / Not-Best Practice ?



<http://knowyourmeme.com/photos/1080551>

000-DTSE-Search-7459-60-DM-5

- ☐ Testing on representative hardware
- ☐ Testing/developing on production sized data
- ☐ Having a ready/representative test (query) harness
- ☐ Having a Rabbit's Foot

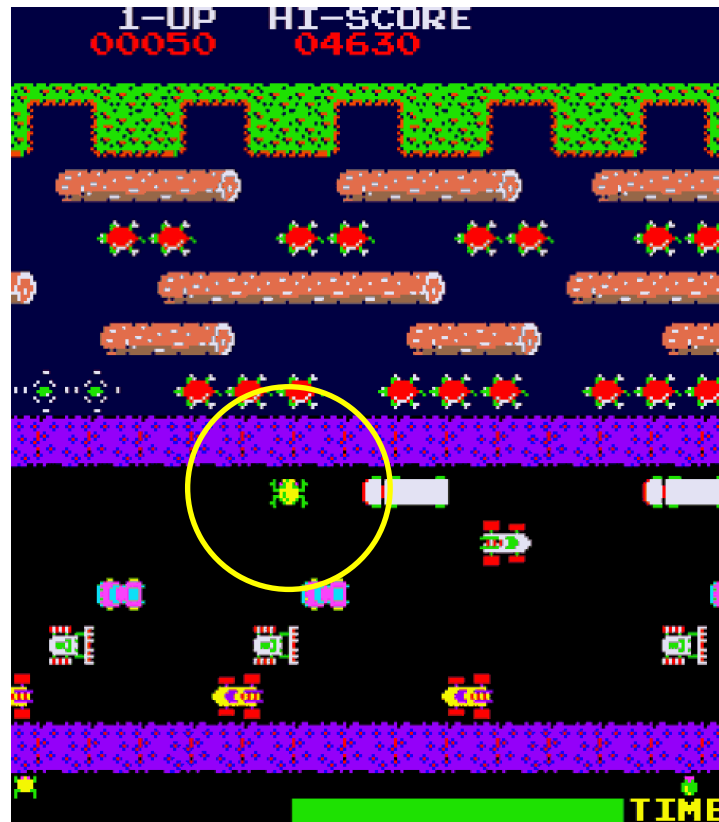
End of Discussion Lab:

DSE Search: Cap Planning, Tuning-

System tuning

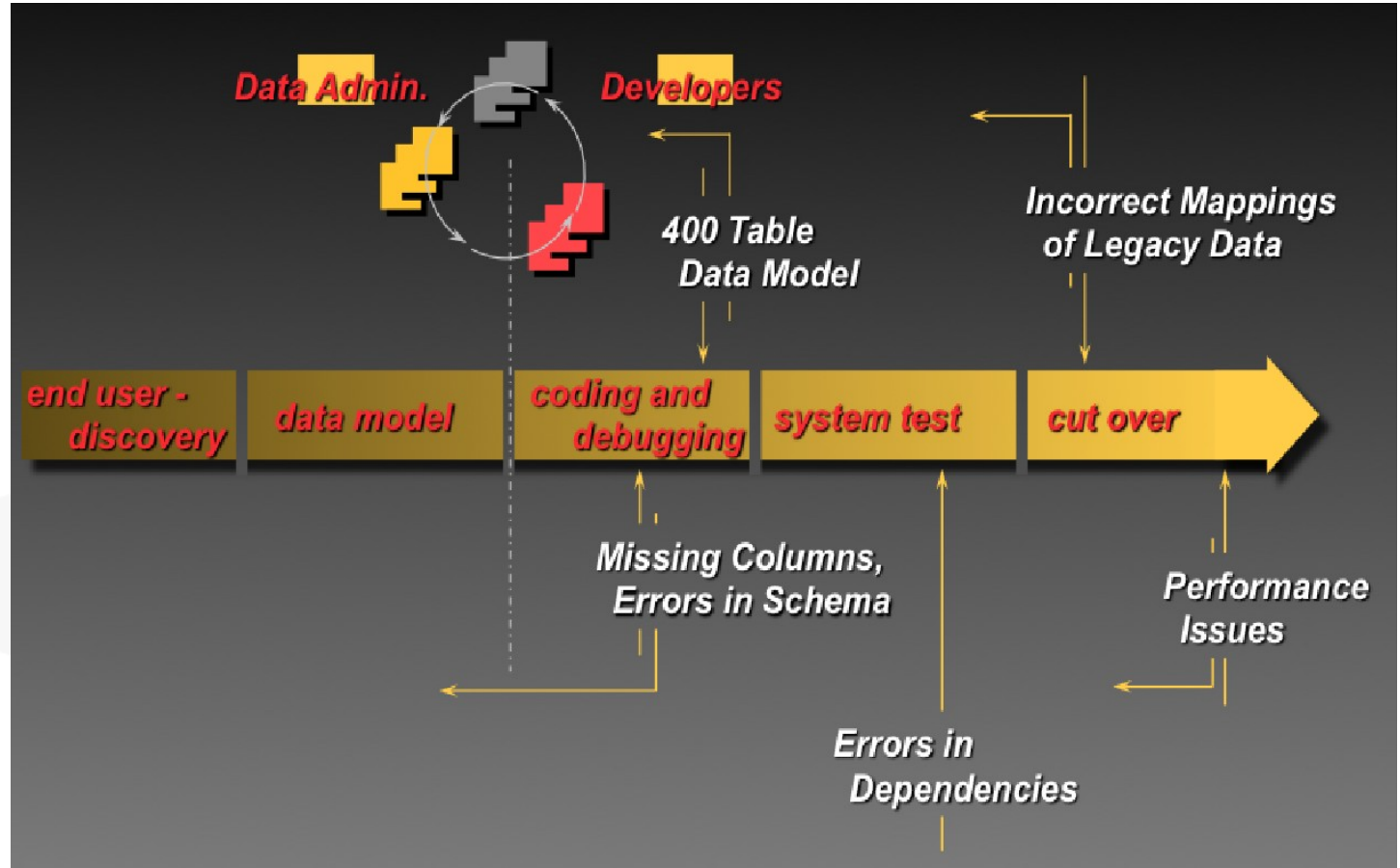
Statement tuning (queries)

- Prod sized data, repeatable test harness, isolation
- Frequency, concurrency, spread
- One variable; rinse, repeat

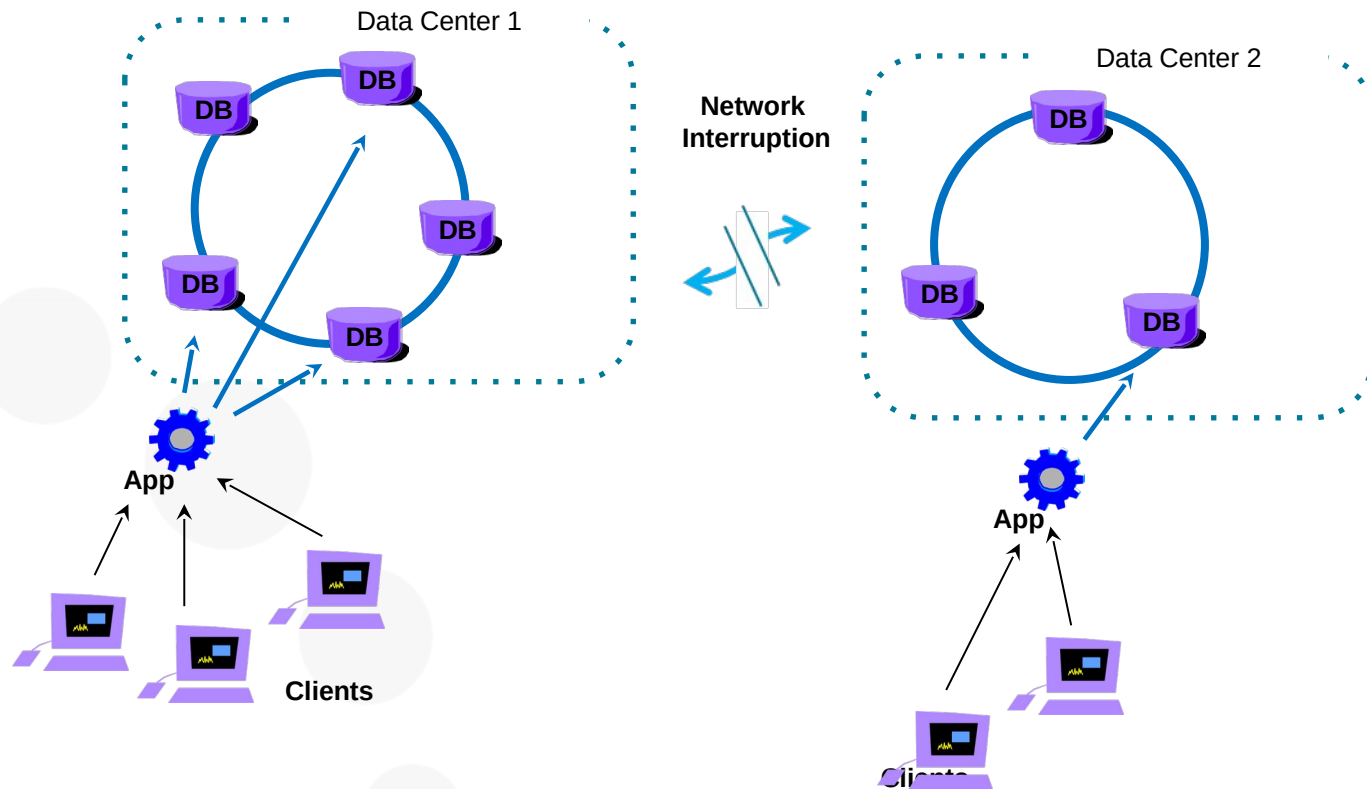


<http://www.classicgaming.cc/classics/frogger/about>

Common SDLC, and Issues



DSE Search: DCs, Read/Write, Load isolation



If you don't have the Query Harness-

1

- Developers
 - This is something someone should have created/maintained
 - Reverse engineering is not forecasting, signing your name to something

2

- CQL Slow Log (dse.yaml. perm)
 - enabled
 - threshold_ms
 - ttl_seconds
 - async_writers
- dsetool perf cqslowlog enable (temp)
- dsetool perf cqslowlog 1000
- ```
SELECT * FROM
dse_perf,node_slow_log;
```

- Solr Slow Sub-query Log
  - `dsetool perf solrslowlog enable`
  - `dsetool perf solrindexstats enable`
  - `dsetool perf solrcachestats enable`
  - `dsetool perf solrlatencysnapshots enable`
  - `SELECT * FROM dse_perf.solr_slow_sub_query_log;`
  - `SELECT` (8-10 others)
- DSE Audit Secure subsystem
  - [https://docs.datastax.com/en/dse/6.0/dse-admin/datastax\\_enterprise/security/secAuditEnable.html](https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secAuditEnable.html)

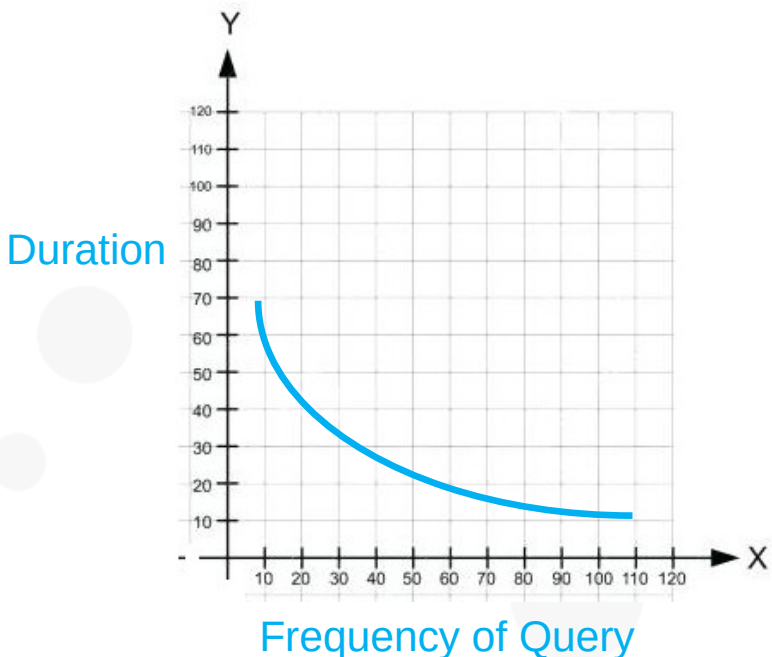
3

## If you don't have the Query Harness-



- DSE Ops Center
  - Search requests, latencies, timeouts, errors
  - Index size

# Start working the (graph)



- Not cost effective to fix every query; Pick the top n% that you have time for
- Index only necessary columns
  - CREATE SEARCH INDEX ON t1;
  - First graph vertex predicate
  - Is it a better DSE Core index or DSE Search index application-

# Start working the (graph)

- CQLSH (driver); tracing on
- Solr Admin UI screen
- CQL statement tuning
  - (Everything/most you know from SQL)
  - Leading wildcards, complex expressions
  - Index negation statements; OR topped
  - Unnecessary sorts
  - Casting
  - If it doesn't reduce network comm, move to the client tier
- Query Metrics Mbean
  - jmxterm, other
  - com.datastax.bdp -> search -> (core name) ->
    - QueryMetrics
    - IndexPool
    - CommitMetrics
    - UpdateMetrics
  - Latency counts, latency percentile, average latency
  - Read path phases; coordinate, execute, retrieve

# “q” and “fq”

- filtercache
  - Easiest/most-common means to OOM
  - dse.yaml – ramBufferSizeMB, ram\_buffer\_heap\_space\_in\_mb
  - solrconfig.xml, highWaterMarkMB=“256”, lowWaterMarkMB=“128”
- dse.yaml (pre 6.0) – max\_solr\_concurrency\_per\_core, back\_pressure\_theshhold\_per\_core



# Call to Action-



<http://picsnook.com/sunset-desert-minimal-4k-hd-wallpaper/>

- We've outlined the knobs-
- It's about changing one variable at a time; careful testing
- Invest in the query harness
  - Remember the incremental issue versus overnight ? Two very different problems.



# End of Module: